

SECURE NETWORK WITH FIREBALLS NET, PORT FILTERING, PACKET FILTERING, AND OTHER METHODS

1. Set up Windows Defender Firewall on your Windows laptop or PC and create a new inbound rule to block all incoming traffic on port 8080. Take a screenshot of the rule you created.

Answer

Create the rule

1. *Press Win + R, type wf.msc, and press Enter.*
2. *Select Inbound Rules.*
3. *Click New Rule... on the right.*
4. *Select Port → Next.*
5. *Select TCP.*

6. *Choose Specific local ports and enter 8080.*
7. *Select Block the connection.*
8. *Select Domain, Private, and Public.*
9. *Name it Block Inbound TCP 8080.*
10. *Click Finish.*

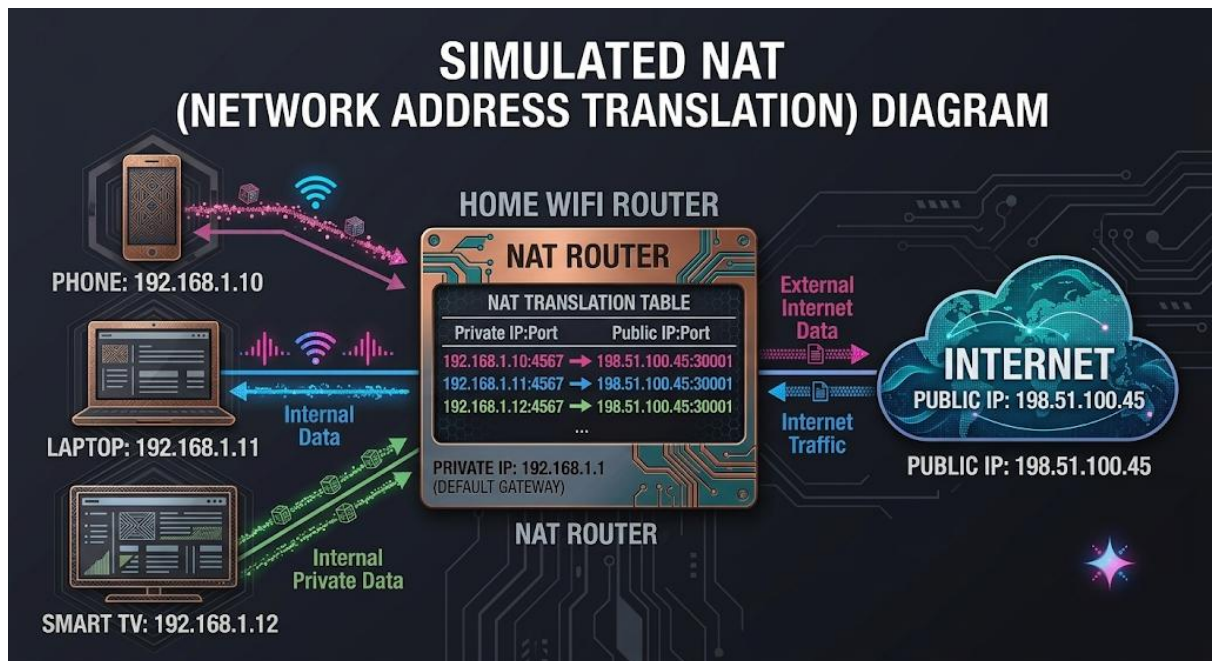
2. Simulate NAT (Network Address Translation) by drawing a simple diagram showing how your home WiFi router assigns private IP addresses to your phone, laptop, and smart TV, and how it translates them to a single public IP for internet access. Use any drawing tool or even hand-draw and click a photo.

Answer

How NAT works in this setup:

- Your phone, laptop, and smart TV all use private IPs.
- The router rewrites (translates) every outgoing request so it appears to come from the single public IP.

- When replies come back, the router translates them back to the correct private device.



3. Configure port filtering using your WiFi router's admin panel (or use a router simulator online if you don't have access). Block outgoing traffic on port 25 (SMTP) and explain in 2 sentences how this could help

prevent spam from your network.

Answer

Port Filtering = Block Outgoing Port 25 (SMTP)

Typical steps on most home routers (TP-Link / D-Link / similar)

- 1. Log in to the router admin page (192.168.1.1 or 192.168.0.1).*
- 2. Go to Advanced → Security → Access Control / Port Filtering / Firewall → Outbound Filter.*
- 3. Create a new rule:*
 - Direction: Outbound / Outgoing*
 - Protocol: TCP*
 - Port: 25*
 - Action: Block / Deny*
 - Apply to: All devices (or LAN)*
- 4. Save / Apply the rule.*

4. Given a sample packet filtering rule set (provided by your trainer), identify which of these three sample packets would be allowed or blocked: 1) HTTP request from 192.168.1.5 to 172.217.166.78:80, 2) SSH request from 10.0.0.10 to 34.201.23.1:22, 3) DNS query from 192.168.1.7 to 8.8.8.8:53. Focus on source/destination IPs and ports in the rule set.

Answer

Packet filtering firewall rules are processed sequentially (usually top-to-bottom), matching packets based on:

- *Source IP / Network*
- *Destination IP / Network*
- *Destination Port / Protocol (e.g., Port 80 for HTTP, Port 22 for SSH, Port 53 for DNS)*
- *Action (ALLOW / PERMIT or BLOCK / DENY)*

Packets Ready for Evaluation:

1. Packet 1 (HTTP):

- *Source IP: 192.168.1.5*
- *Destination IP: 172.217.166.78*
- *Destination Port: 80 (TCP/HTTP)*

2. Packet 2 (SSH):

- *Source IP: 10.0.0.10*
- *Destination IP: 34.201.23.1*
- *Destination Port: 22 (TCP/SSH)*

3. Packet 3 (DNS):

- *Source IP: 192.168.1.7*
- *Destination IP: 8.8.8.8*
- *Destination Port: 53 (UDP/DNS)*

5. Research and list 3 ways Instagram or Zomato might use firewalls and packet filtering to protect their servers from attacks or misuse.

Answer

1. Edge / Perimeter Packet Filtering with IP, Port, and Protocol Rules (ACLs)

Both companies place firewalls (or software-defined equivalents) at the network edge (PoPs, data-center

boundaries, or cloud ingress points) that inspect every incoming packet's header.

- Rules allow only expected traffic: e.g., TCP port 443 (HTTPS) to public web/API endpoints, specific ports for internal services, and drop everything else by default.
- Source/destination IP filtering blocks known malicious ranges, spoofed addresses, or traffic from regions/ASNs that should not reach production servers.
- Meta runs high-performance BPF-based edge firewalls (and has discussed managing tens of thousands of ACL rules containing hundreds of thousands of addresses) that perform these lookups efficiently even under high load. This also prevents “DoS on closed ports” by dropping unwanted packets early, before they consume resources. Zomato would use the same approach to protect order, payment, and restaurant-partner APIs—only permitting legitimate client or partner traffic.

2. Stateful Inspection and Connection Tracking

Rather than treating every packet in isolation, modern firewalls maintain a state table of active connections.

- When a legitimate client initiates a connection (e.g., an Instagram app or Zomato user opening a session), the firewall records the 5-tuple (source/destination IP + ports

+ protocol). Return traffic belonging to that established session is allowed; unsolicited inbound packets that do not match any state are dropped.

- This stops many reconnaissance scans, unauthorized connection attempts, and certain flooding attacks while still permitting normal bidirectional traffic.*
- Meta's public discussions of its firewalls emphasize stateful behavior and the ability to chain additional logic (rate limiting, custom filters) on top of the basic state checks.*

3. Rate Limiting, Anomaly Detection, and Targeted Dropping of Abusive Packets

Packet-filtering rules can enforce thresholds and drop or throttle traffic that exceeds them.

- Limit the number of new connections (or packets) per source IP in a short time window. Exceeding the limit causes subsequent packets to be dropped—effective against brute-force login attempts, credential stuffing, simple volumetric DDoS, and API abuse (e.g., scraping or order-spam on Zomato).*
- Additional filters can drop malformed packets, unexpected protocols, or traffic matching known attack signatures*

(Meta has referenced Suricata-compatible rules and AWS Network Firewall for this purpose in cloud environments).

- *Egress filtering can also be applied: servers are only allowed to initiate outbound connections to approved destinations, reducing the risk of data exfiltration if a host is compromised.*